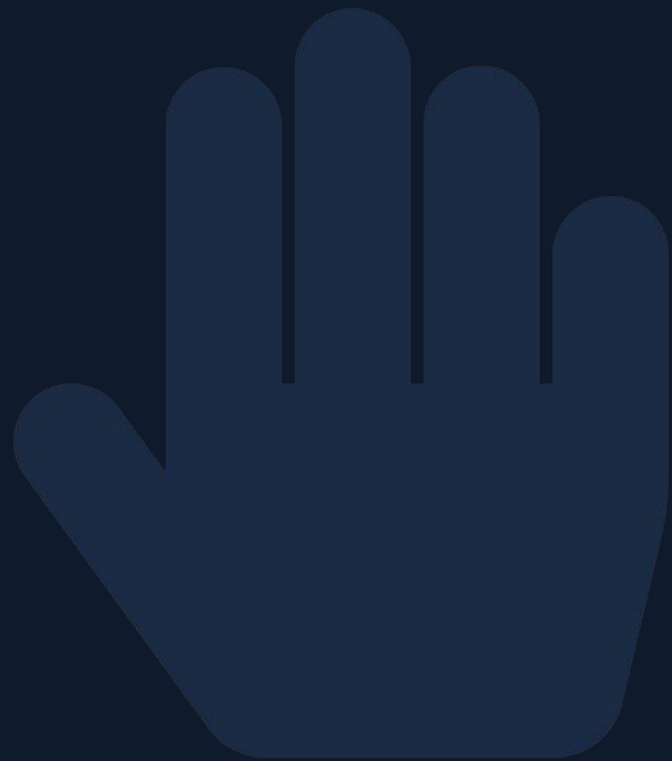


A MEDTOUREASY TRAINEESHIP PROJECT REPORT

Do left-handers really die young?

A Bayesian look at a famous 9-year gap, and the people, history and data behind it.

Prepared by Ansuman · Python, pandas, NumPy, Matplotlib



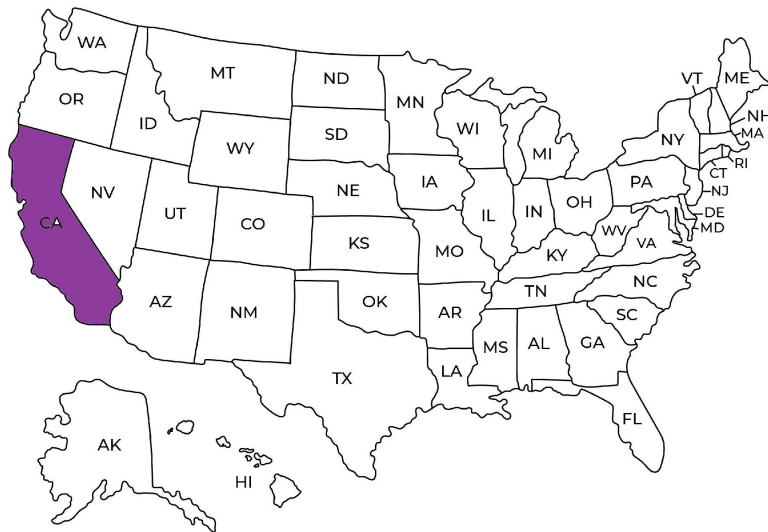
In 1991, a study claimed left-handers die 9 years sooner

“

Left-handers died at an average of **66**. Right-handers at **75**.

Halpern and Coren wrote to families of people who died in 1989 in two southern California counties and asked one question: which hand did your relative use?

They blamed correlates of left-handedness and accidents in a right-handed world.



Original study: two counties in southern California

A gap that big should have been impossible to miss

01

Bigger than smoking

Critics pointed out the gap was larger than the one between smokers and non-smokers.

02

Insurers never saw it

Actuaries price risk for a living. A 9-year effect would have shown up in their tables.

03

Society had changed

Left-handed children were once pushed to switch hands, so older people rarely reported it.

OUR QUESTION Can the changing rate of left-handedness alone reproduce the gap, with no real difference in lifespan?

Project objectives and core deliverables



Map the pattern

Measure how reported left-handedness changes with age and birth year in the 1986 survey.

DELIVERABLE: TASKS 1 TO 2



Model with Bayes

Turn $P(\text{left-handed} \mid \text{age at death})$ into $P(\text{age at death} \mid \text{left-handed})$ for both groups.

DELIVERABLE: TASKS 3 TO 8



Test the claim

Compare mean ages, rerun for 2018, and check the result against published research.

DELIVERABLE: TASKS 9 TO 10

Our handedness data came from a smell survey



1,177,507

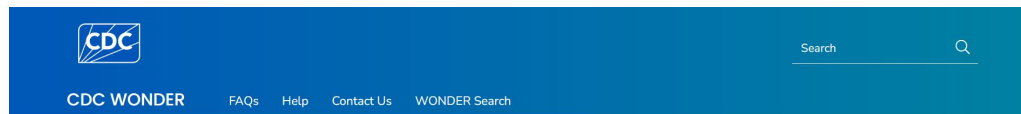
Americans aged 10 to 86 answered questions on writing and throwing hand in 1986.

The questions rode along on the National Geographic Smell Survey, run with the Society and its magazine by the Monell Chemical Senses Center.



Charles J. Wysocki
Avery N. Gilbert

Every US death certificate from one year: 1999



Underlying Cause of Death 1999-2020

Summary: The Underlying Cause of Death data available on WONDER are county-level national mortality and population data spanning the years 1999-2020. Data are based on death certificates for U.S. residents. Each death certificate identifies a single underlying cause of death and demographic data. The number of deaths, crude death rates or age-adjusted death rates, and 95% confidence intervals and standard errors for death rates can be obtained by place of residence (total U.S., region, state and county), age group (single-year-of age, 5-year age groups, 10-year age groups and infant age groups), race, Hispanic ethnicity, sex, year, cause-of-death (4-digit ICD-10 code or group of codes), injury intent and injury mechanism, drug/alcohol induced causes and urbanization categories. Data are also available for place of death, month and week day of death, and whether an autopsy was performed.

Privacy policy: Statistics representing fewer than ten (one to nine) deaths or births are suppressed. Population figures are also suppressed when the population represents fewer than ten persons. Additional privacy constraints apply to infant mortality statistics for infant age groups and live births denominator population figures. See [Assurance of Confidentiality](#) for more information.

Mortality data: The mortality data are based on information from all death certificates filed in the fifty states and the District of Columbia. Deaths of nonresidents (e.g. nonresident aliens, nationals living abroad, residents of Puerto Rico, Guam, the Virgin Islands, and other territories of the U.S.) and fetal deaths are excluded. Mortality data from the death certificates are coded by the states and provided to NCHS through the Vital Statistics Cooperative Program or coded by NCHS from copies of the original death certificates provided to NCHS by the State registration offices. For more information, see [Technical Appendix from Vital Statistics of United States: 1999 Mortality](#).

- **About suppressed data and unreliable rates:**

2,391,399

deaths in the US in 1999, counted by single year of age

76.7

years life expectancy at birth that year

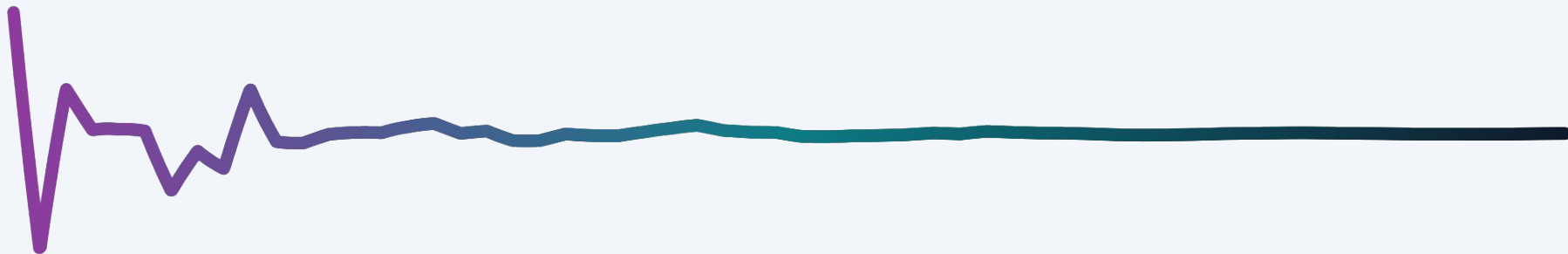
0 to 120

the age range covered, one row per age

This gives $P(A)$: how likely a person is to die at each age.

CDC mortality data are built from all death certificates filed in the 50 states and DC, with one underlying cause per death.

From two noisy snapshots to one clear answer



COLLECT

Survey CSV and CDC TSV loaded with pandas; tab separator and skipped header row.

CLEAN

Dropped missing death counts; built Birth_year and Mean_lh columns.

MODEL

Rates by birth year, flat averages outside ages 10 to 86, then Bayes for both groups.

One formula flips the question around

$$P(A \mid LH) = P(LH \mid A) \times P(A) \div P(LH)$$

$P(A \mid LH)$

WHAT WE WANT

Chance a left-hander died at age A.

$P(LH \mid A)$

FROM THE 1986 SURVEY

Chance someone who died at A was left-handed, based on their birth year.

$P(A)$

FROM 1999 DEATHS

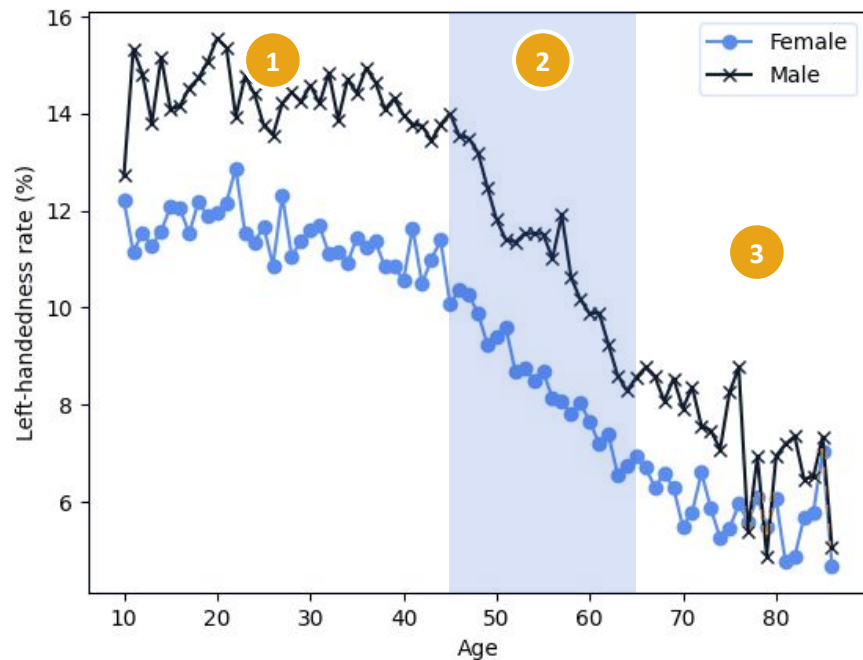
Chance anyone died at age A.

$P(LH)$

WEIGHTED AVERAGE

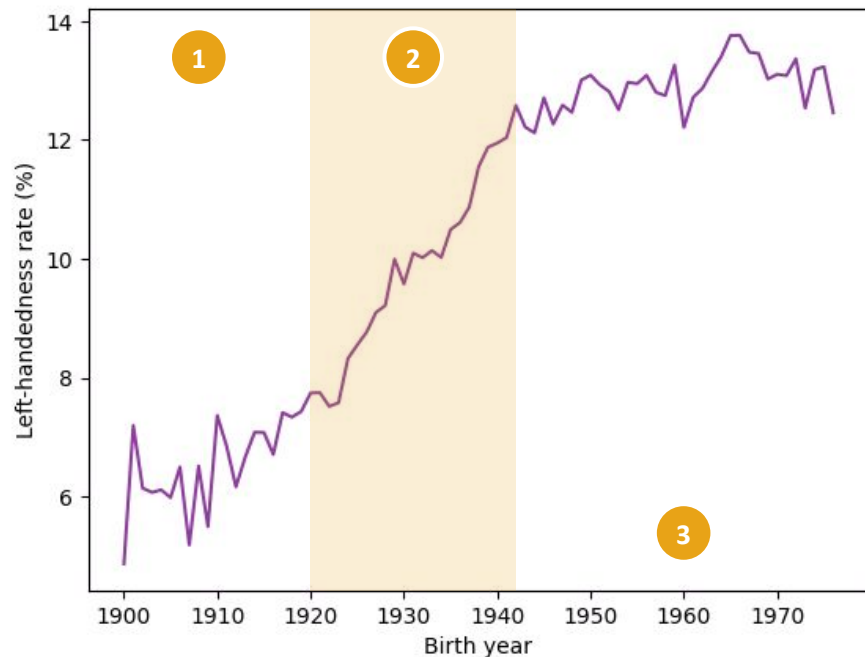
Share of left-handers among all the dead, about 7.8%.

Older Americans report far less left-handedness



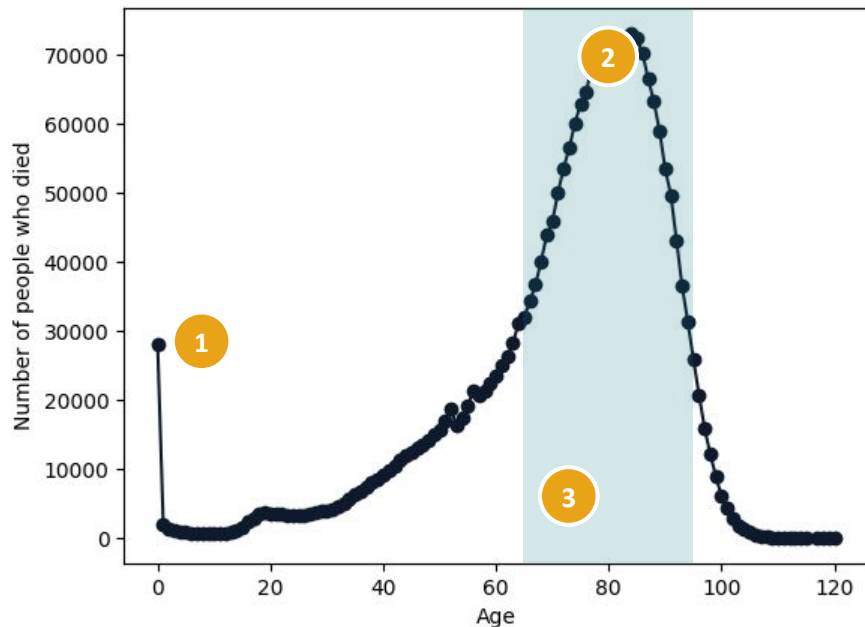
- 1 Men lead at every age**
About 14 to 15.5% of men under 40 vs 11 to 12% of women.
- 2 A cliff from 45 to 65**
Rates fall about 40% in 20 years. These people were born roughly 1921 to 1941.
- 3 Oldest ages are noisy**
Fewer respondents past 70, so points zig-zag. The model leans on them later.

Birth year matters, not how old you are



- 1 About 6% before 1920**
Flat and low for the earliest generations.
- 2 A steady 20-year climb**
From 7.5% to 12% between about 1920 and 1942. Gradual, like attitudes relaxing school by school.
- 3 Plateau near 13%**
Close to today's worldwide estimate of 10.6%. This is roughly the natural rate.

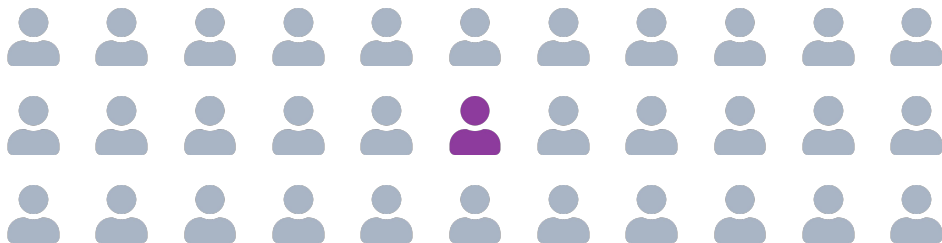
The dead come mostly from low left-handed generations



- 1 Infant spike**
About 28,000 deaths at age 0, then childhood is the safest stretch of life.
- 2 Peak around 84**
Counts, not risk: many people still alive times a high yearly risk.
- 3 Born before about 1925**
In a 1990 study year, deaths at 65 to 95 (shaded) belong to people born around 1895 to 1925.

How a gap appears when nobody dies early

Older generation dies at 80 · 3% report left-handed



Younger generation dies at 40 · 12% report left-handed



Illustration only. Each figure is a death. Both generations have identical lifespans.

DEAD LEFT-HANDERS

60

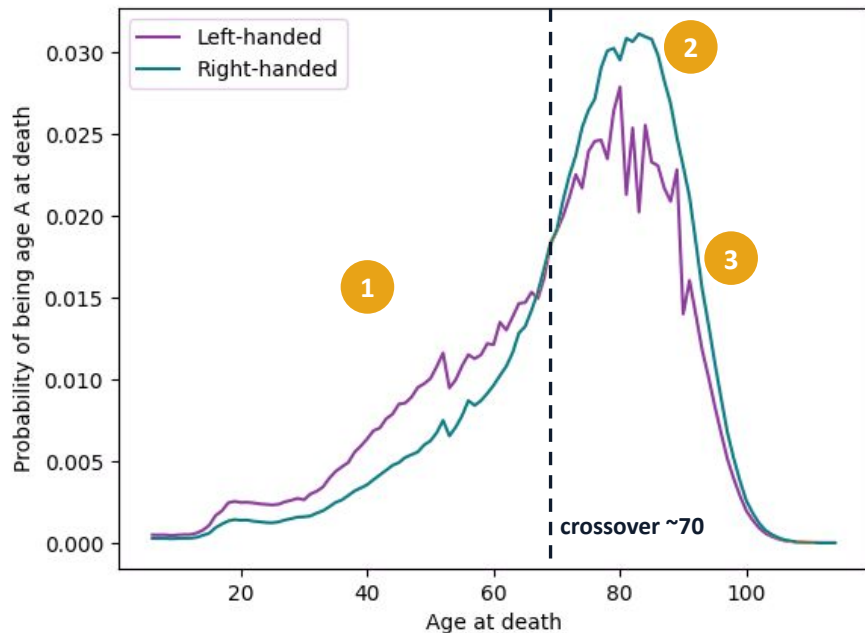
average age. Half of them are young.

DEAD RIGHT-HANDERS

73

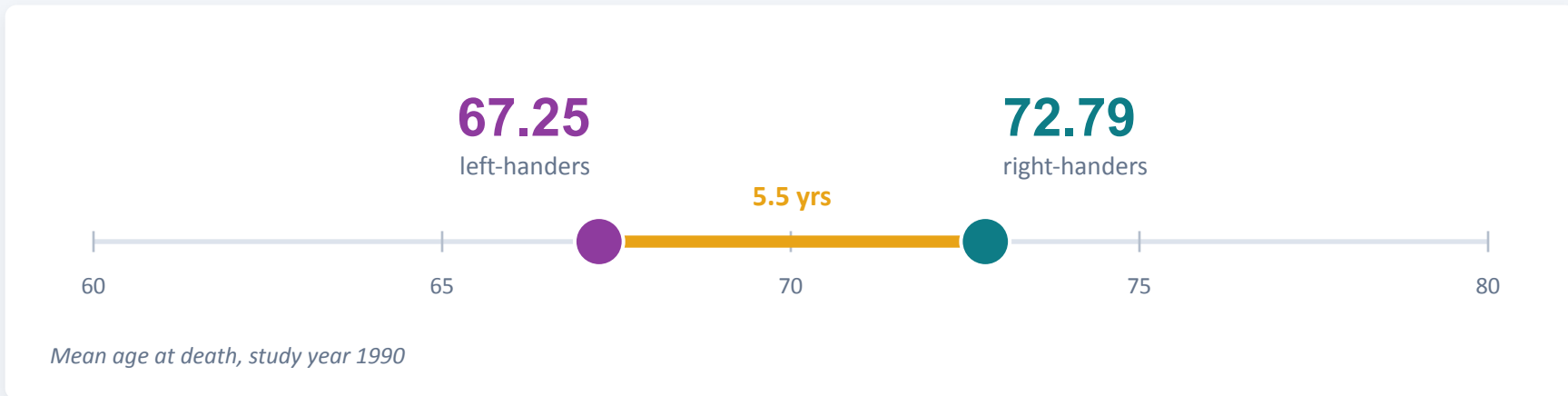
average age. Mostly the old generation.

Left-handers skew younger among the dead



- 1 Above until about 70**
Around 40, left-handers are 1.5 times as likely to appear. Around 20, closer to 1.8 times.
- 2 Right-handers peak later**
About age 83, versus about 80 for left-handers.
- 3 A step at 90 is the model**
Beyond age 90 the code switches to a flat early-1900s rate. Not real deaths.

Reporting alone creates a 5.5-year gap



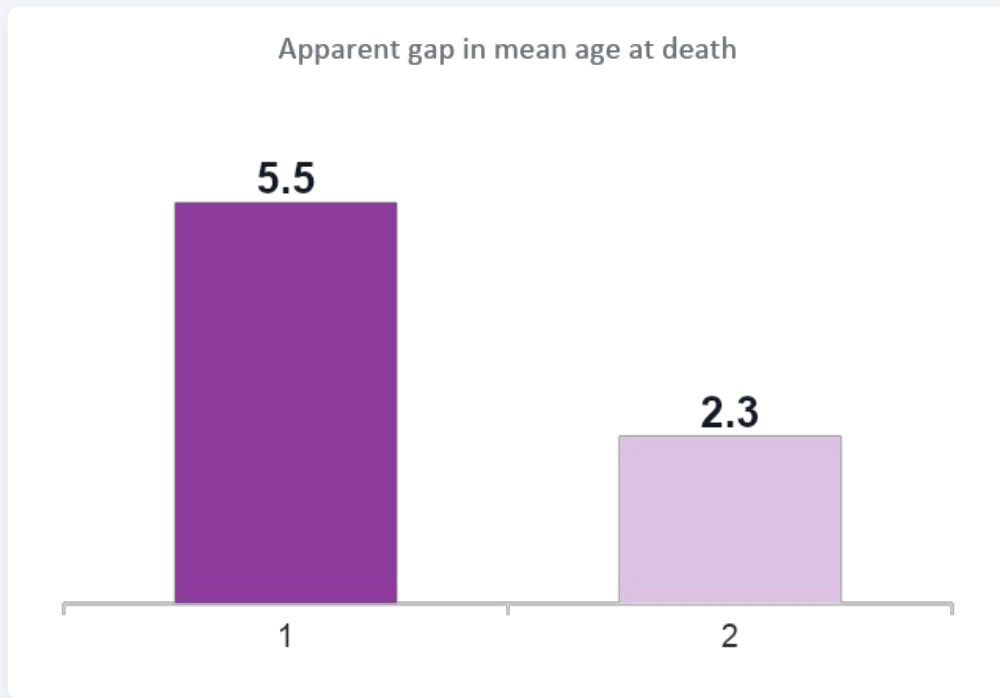
60%

of the original 9-year gap is reproduced with zero lifespan difference.

66 vs 67.25

Our left-handers land close to the original study. Most of the shortfall sits on the right-handed side.

Move the study to 2018 and the gap fades

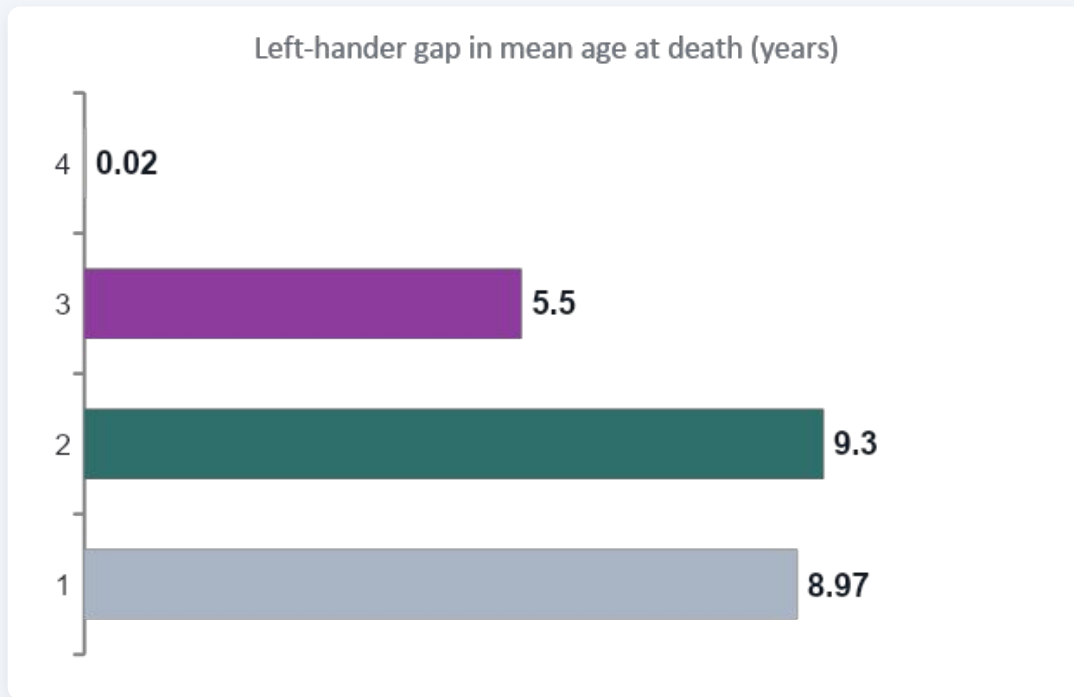


The gap falls by 58%

- People dying in 2018 at 80 were born in 1938, near the top of the climb, so left-handedness was reported more evenly among them.
- A real biological effect would not fade like this. An artefact does.
- It should approach zero once deaths come mostly from people born after 1942, roughly the 2030s to 2040s.

What-if only: the 2018 run reuses the 1999 death pattern.

A 2023 study confirms the whole gap is an artefact



- 1 9.3 years with no lifespan effect**
Microsoft AI for Good and Harvard modelled all 2,019,512 US deaths of 1989.
- 2 0.02 years if the rate never changed**
Take away the reporting shift and the gap vanishes.
- 3 Why we get 5.5**
1999 data, whole US, sexes combined, extrapolated ages.

What limits this model, and what comes next

Wrong year of deaths

1999 deaths used for a 1990 study year, a 9-year shift in birth years.

Wrong place

Whole US, not two California counties.

Sexes combined

Men are more left-handed and die younger on average.

Edges are guessed

Ages under 14 and over 90 use flat averages of the noisiest survey points.

NEXT STEPS

- Rerun by sex using the separate male and female columns.
- Sweep every study year to chart when the gap disappears.
- Hold the rate constant to prove the gap goes to zero.
- Pull 1989 deaths from CDC data to match the original year.

CONCLUSION

Left-handers were not dying young. They were being counted differently.

5.5

years of gap from reporting alone

60%

of the famous 9 years explained

2.3

years left in a 2018 what-if

For MedTourEasy: before trusting any health statistic, ask who was counted, when, and how. Changing social norms can create patterns that look like biology.

EVIDENCE USED IN THIS BRIEFING

Halpern, D. F. & Coren, S. (1991). Handedness and life span. The original 9-year claim.

Gilbert, A. N. & Wysocki, C. J. (1992). Hand preference and age in the United States. *Neuropsychologia*, 30(7), 601–608.

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CDC / NCHS (2001). Deaths: Final Data for 1999. *National Vital Statistics Reports*, 49(8).

Ferres, J. L. et al. (2023). Modeling to explore and challenge inherent assumptions when cultural norms have changed. *Archives of Public Health*, 81:137.

Papadatou-Pastou, M. et al. (2020). Human handedness: a meta-analysis. *Psychological Bulletin*, 146(6).

MedTourEasy Training & Development Team. Project proposal and datasets.